

Predicting MCI to AD Conversion Using Machine Learning

PHB 243b Project 2 Report

[Your Name Here]

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1 Abstract

Background: Mild Cognitive Impairment (MCI) often precedes Alzheimer’s disease (AD), but not all MCI patients progress to dementia. Identifying individuals at high risk for conversion is critical for clinical management and trial enrollment.

Objective: To develop and validate machine learning models for predicting 3-year conversion from MCI to AD using baseline clinical and cognitive data.

Methods: [Describe your methods in 2-3 sentences]

Results: [Report key findings: sample size, best model AUROC, top predictors]

Conclusion: [Summarize clinical implications in 1-2 sentences]

Keywords: Alzheimer’s disease, mild cognitive impairment, machine learning, prediction, ADNI

2 Introduction

2.1 Background

Alzheimer’s disease (AD) is the most common cause of dementia, affecting an estimated 6.7 million Americans aged 65 and older (Grassi et al., 2019). The disease is characterized by progressive cognitive decline, ultimately leading to loss of independence and death. Mild Cognitive Impairment (MCI) represents a transitional state between normal aging and dementia, with annual conversion rates to AD of approximately 10-15% (Petersen, 2004).

[Your expanded background here - 2-3 paragraphs]

2.2 Rationale

Grassi et al. (2019) demonstrated that machine learning algorithms using only clinically-available measures (sociodemographic characteristics, clinical scales, and neuropsychological tests) could predict MCI-to-AD conversion with an AUROC of 0.88 (Grassi et al., 2019). This approach prioritizes clinical translatability over maximum predictive accuracy, excluding expensive imaging and genetic biomarkers.

2.3 Objectives

The primary objective of this study was to replicate and extend the approach of Grassi et al. (2019) using data from the Alzheimer’s Disease Neuroimaging Initiative (ADNI).

Specific aims:

1. Develop machine learning models to predict 3-year MCI-to-AD conversion
2. Compare performance across multiple algorithms
3. Identify the most important predictive features

3 Methods

3.1 Data Source

Data were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) via the AD-NIMERGE2 R package (Donohue et al., 2026). ADNI is a longitudinal multicenter study designed to develop clinical, imaging, genetic, and biochemical biomarkers for early detection and tracking of Alzheimer’s disease. Data were downloaded on [DATE].

3.2 Study Population

3.2.1 Inclusion Criteria

1. Baseline diagnosis of Mild Cognitive Impairment (EMCI or LMCI)
2. At least one follow-up visit with diagnostic assessment
3. Complete data on primary predictor variables

3.2.2 Exclusion Criteria

1. Baseline diagnosis of Cognitively Normal (CN) or Dementia
2. No follow-up visits after baseline

3.3 Outcome Definition

The primary outcome was conversion from MCI to AD dementia, defined as any follow-up diagnosis of “Dementia” within 3 years of baseline assessment. Subjects who remained MCI throughout follow-up or reverted to CN were classified as stable (sMCI). Subjects who progressed to dementia were classified as converters (cAD).

3.4 Predictor Variables

Following Grassi et al. (2019), we included predictor variables from four categories:

Sociodemographic: Age, years of education, sex, marital status

Clinical scales: Clinical Dementia Rating Sum of Boxes (CDR-SB), Functional Activities Questionnaire (FAQ)

Neuropsychological tests: Mini-Mental State Examination (MMSE), Alzheimer’s Disease Assessment Scale-Cognitive 13-item (ADAS-Cog-13), Rey Auditory Verbal Learning Test (RAVLT) immediate recall, learning, and forgetting scores, Logical Memory delayed recall (LDELTOTAL), Trail Making Test Part B (TMT-B)

MCI subtype: Early MCI (EMCI) vs. Late MCI (LMCI)

Imaging and genetic variables were intentionally excluded to maximize clinical translatability.

3.5 Statistical Analysis

3.5.1 Descriptive Statistics

Baseline characteristics were summarized by conversion status using means and standard deviations for continuous variables and frequencies and percentages for categorical variables. Group comparisons used t-tests for continuous variables and chi-square tests for categorical variables.

3.5.2 Machine Learning Models

We implemented three classification algorithms:

1. **Logistic Regression:** Baseline linear classifier with L2 regularization
2. **Random Forest:** Ensemble of decision trees with 500 trees
3. **Gradient Boosting (XGBoost):** Sequential tree ensemble

3.5.3 Model Training and Validation

The dataset was split into training (70%) and test (30%) sets using stratified random sampling to preserve the proportion of converters. Hyperparameter tuning was performed using 5-fold cross-validation within the training set. Final model performance was evaluated on the held-out test set.

3.5.4 Evaluation Metrics

Model discrimination was assessed using the area under the receiver operating characteristic curve (AUROC). Additional metrics included sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV). The optimal classification threshold was determined using Youden’s J statistic.

3.5.5 Feature Importance

Variable importance was assessed using:

- Logistic regression: Standardized coefficients
- Random Forest: Mean decrease in Gini impurity
- XGBoost: Gain-based importance

3.5.6 Software

All analyses were conducted in R version 4.4.x using the following packages: tidyverse, caret, randomForest, xgboost, pROC, and ADNIMERGE2.

4 Results

4.1 Study Population

[Your results description here]

4.2 Model Performance

[Your model performance results here]

4.3 Feature Importance

[Your feature importance results here]

4.4 Sensitivity Analyses

[Your sensitivity analysis results here]

5 Discussion

5.1 Principal Findings

[Your principal findings here]

5.2 Comparison with Prior Work

Our findings are [consistent with / different from] those of Grassi et al. (2019), who reported an AUROC of 0.88 using a similar approach. [Discuss similarities and differences in performance, feature importance, etc.]

[Your comparison discussion here]

5.3 Clinical Implications

[Your clinical implications discussion here]

5.4 Strengths and Limitations

Strengths:

1. Use of well-characterized ADNI cohort with standardized assessments
2. Focus on clinically-available predictors for translatability
3. Comparison of multiple machine learning algorithms
4. Rigorous cross-validation methodology

Limitations:

1. ADNI population may not generalize to community settings
2. Retrospective design cannot establish causation
3. No external validation cohort
4. [Additional limitations specific to your analysis]

5.5 Future Directions

[Your future directions here]

6 Conclusion

[Your conclusion here]

7 References

- Donohue, M. C., Hussen, K., Langford, O., Gallardo, R., Jimenez-Maggiora, G., & Aisen, P. S. (2026). Alzheimer's clinical research data via r packages: The alzverse. *Alzheimer's & Dementia*, 22, e71152. <https://doi.org/10.1002/alz.71152>
- Grassi, M., Rouleaux, N., Caldirola, D., Loewenstein, D., Schruers, K., Perna, G., & Dumontier, M. (2019). A novel ensemble-based machine learning algorithm to predict the conversion from mild cognitive impairment to alzheimer's disease using socio-demographic characteristics, clinical information, and neuropsychological measures. *Frontiers in Neurology*, 10, 1–15. <https://doi.org/10.3389/fneur.2019.00756>
- Petersen, R. C. (2004). Mild cognitive impairment as a diagnostic entity. *Journal of Internal Medicine*, 256(3), 183–194. <https://doi.org/10.1111/j.1365-2796.2004.01388.x>

8 Supplementary Material

8.1 Supplementary Table S1: Complete Model Coefficients

8.2 Supplementary Figure S1: Calibration Plots

8.3 Supplementary Figure S2: Learning Curves

9 Session Information

```
sessionInfo()
```

```
R version 4.6.1 (2026-06-24)
Platform: aarch64-apple-darwin25.4.0
Running under: macOS Tahoe 26.5.1

Matrix products: default
BLAS: /opt/homebrew/Cellar/openblas/0.3.34/lib/libopenblas-r0.3.34.dylib
LAPACK: /opt/homebrew/Cellar/r/4.6.1/lib/R/lib/libRlapack.dylib; LAPACK version 3.12.1

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time zone: America/Los_Angeles
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attached base packages:
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other attached packages:
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[9] purrr_1.2.2 readr_2.2.0 tidyr_1.3.2 tibble_3.3.1
[13] ggplot2_4.0.3 tidyverse_2.0.0
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loaded via a namespace (and not attached):

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